theme of this chapter, we'll get started by creating a big dataset of random points and render them in SVG.

### 11.3.1 Generating random xy data

Our third random data generator doesn't require nearly as much work as the first two did. In the following listing, all we do is create 3000 points with random x and y coordinates.

**Listing 11.13 xy data generator**

```
sampleData = d3.range(3000).map(function(d) {
  var datapoint = {};
  datapoint.id = "Sample Node " + d;
  datapoint.x = Math.random() * 500;
  datapoint.y = Math.random() * 500;

  return datapoint;
})

d3.select("svg").selectAll("circle")
  .data(sampleData)
  .enter()
  .append("circle")
  .attr("class", "xy")
  .attr("r", 3)
  .attr("cx", function(d) {return d.x})
  .attr("cy", function(d) {return d.y});
```

**Because we know the fixed size of our canvas, we can hardwire this.**

As you may expect, the result of this code, shown in figure 11.11, is a bunch of pink circles scattered randomly all over our canvas.

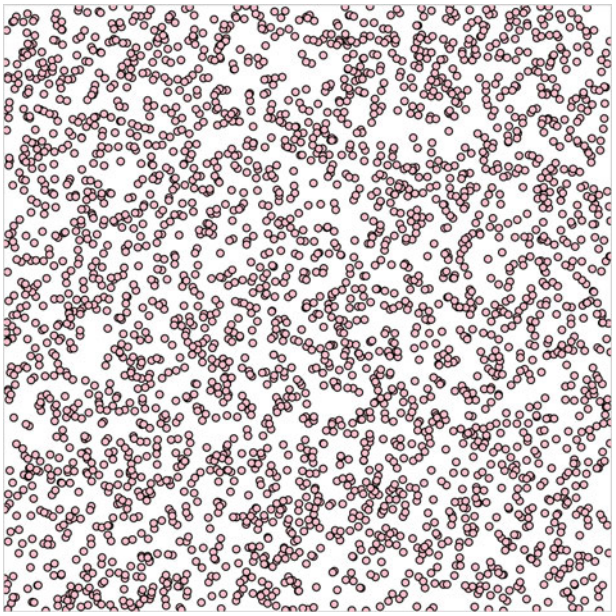


**Figure 11.11 3000 randomly placed points represented by pink SVG `<circle>` elements**